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Thermodynamic description of the Pb–Yb binary system

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ABSTRACT

Thermodynamic modelling of the Pb–Yb binary system was carried out with the help of the CALPHAD method. The liquid phase has been described with the association solution model with 'Pb_1Yb_2' as an associated complex. The solution phases BCC_A2 and FCC_A1 were modelled with the sublattice formalism. The αPbYb_LT and βPbYb_HT Pb sub-stoichiometric intermetallic compounds, which have a homogeneity range, were treated with the formula $(Pb, Yb)_{0.5}(Yb)_{0.5}$ by a two-sublattice model with Pb and Yb on the first sublattice and Yb on the second one. Pb_3Yb, Pb_3Yb_5 and $PbYb_2$ have been treated as stoichiometric compounds. The calculations based on the thermodynamic modelling are in good agreement with the phase diagram data and experimental thermodynamic values.

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1. Introduction

This study is a part of a thermodynamic investigation of M–Pb systems (M = Ba [1], Ca [2]) which is intended to give a better understanding of the constitutional properties and potential technological applications of these alloys.

This present work deals with an assessment of the thermodynamic description of the Pb–Yb system using the CALPHAD technique [3]. The thermodynamic parameters involved in the models are optimized from the experimental thermodynamic and phase diagram data.

2. Review of experimental data

The currently accepted Pb–Yb phase diagram is based on an investigation carried out by McMasters and Gschneidner [4]. Four intermetallic compounds were described: Pb_3Yb, PbYb and $PbYb_2$, which melt congruently at 1013 K, 1383 K and 1519 K, respectively; while Pb_3Yb_5 is a compound formed by a peritectic reaction at 1423 K (Fig. 1). Borzone et al. [5] examined Pb–Yb alloys by X-ray diffraction and metallographic analyses; the crystal structures of the different compounds were confirmed. They observed a homogeneity range from 50 to 52 at.% Yb for PbYb in the 573–823 K temperature range and negligible solubility ranges for the other compounds. In addition, a polymorphic transformation was reported for the PbYb compound around 773 K and the crystal structure of the high temperature modification was determined. Three terminal solid solutions exist: (Pb), (γYb) and (βYb). The solubility of Yb in (Pb) is very low and estimated to be lower than 0.15 at.% Yb at room temperature. In the Yb-rich region, the solubility of Pb in (βYb) is negligible and can be fixed at 0.18 at.% Pb at room temperature. The crystal structures of the various phases are reported in Table 1. In the range 323 to 823 K, [5] obtained C_p values of $PbYb_2$, PbYb and Pb_3Yb by differential scanning calorimetry. A new phase, $Pb_{20}Yb_{31}$, has been identified, and its crystal structure was determined by Ganguli et al. [6].

The enthalpy of formation of the intermetallic compounds has been measured by Schiffman et al. [7] using the Knudsen data method. Recently, Sommer et al. [8] examined Pb–Yb alloys by X-ray diffraction. Sommer et al. [8] determined the enthalpies of formation of Pb–Yb liquid and solid alloys. The nominal purities of the metals were Pb (99.999 mass.%) and Yb (99.98 mass.%) and the Pb–Yb alloys were examined by X-ray diffraction. The enthalpies of formation of Pb–Yb solid alloys at 300 K have been determined by direct calorimetry (with an aneroid isoperibol calorimeter). The enthalpies of mixing of liquid have been measured in the 960–1100 K temperature range by using an isoperibolic high temperature solution calorimetry. The heat of formation and heat of melting for the Pb_3Yb compound have been reported by Palenzona et al. [9].

3. Thermodynamic models

3.1. Pure elements

- The Gibbs energy function $G_i^{\phi}(T) = {}^0G_i^{\phi} - H_i^{SER}$ (298.15 K) for the element i (i = Pb, Yb) in the phase ϕ (ϕ = Liquid, BCC_A2 or FCC_A1) is described by an equation of the following form:

$$G_i^{\phi}(T) = a + bT + cT\ln T + dT^2 + eT^3 + fT^7 + gT^{-1} + hT^{-9} \quad (1)$$

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Table 1
Symbols and crystal structures of the stable solid phases in the (Pb–Yb) alloys.

Diagram symbol	Composition at.% Yb	Symbol used in thermo-calc data file	Pearson symbol	Space group	Strukturbericht symbol	Prototype
(αYb)	100	HCP_A3	$hP2$	$P6_3/mmc$	$A3$	Mg
(βYb)	100	FCC_A1	$cF4$	$Fm\bar{3}m$	$A1$	Cu
(γYb)	100	BCC_A2	$cI2$	$Im\bar{3}m$	$A2$	W
(Pb)	0	FCC_A1	$cf4$	$Fm\bar{3}m$	$A1$	Cu
$PbYb_2$	66.7	$PbYb_2$	$oP12$	$Pnma$	$C23$	Co_2Si
Pb_3Yb_5	62.5	Pb_3Yb_5	$hP16$	$P6_3/mcm$	$D8_8$	Mn_5Si_3
$Pb_{20}Yb_{31}$	60.78	Not optimized in the present work	$tI204$	–	–	$Pu_{31}Pt_{20}$
αPbYb	50	PbYb_LT	$tP2$	$P4/mmm$	$L1_0$	AuCu–I
βPbYb	50	PbYb_HT	$cI2$	$Im\bar{3}m$	$A2$	W
Pb_3Yb	25	Pb_3Yb	$cP4$	$Pm\bar{3}m$	$L1_2$	$AuCu_3$

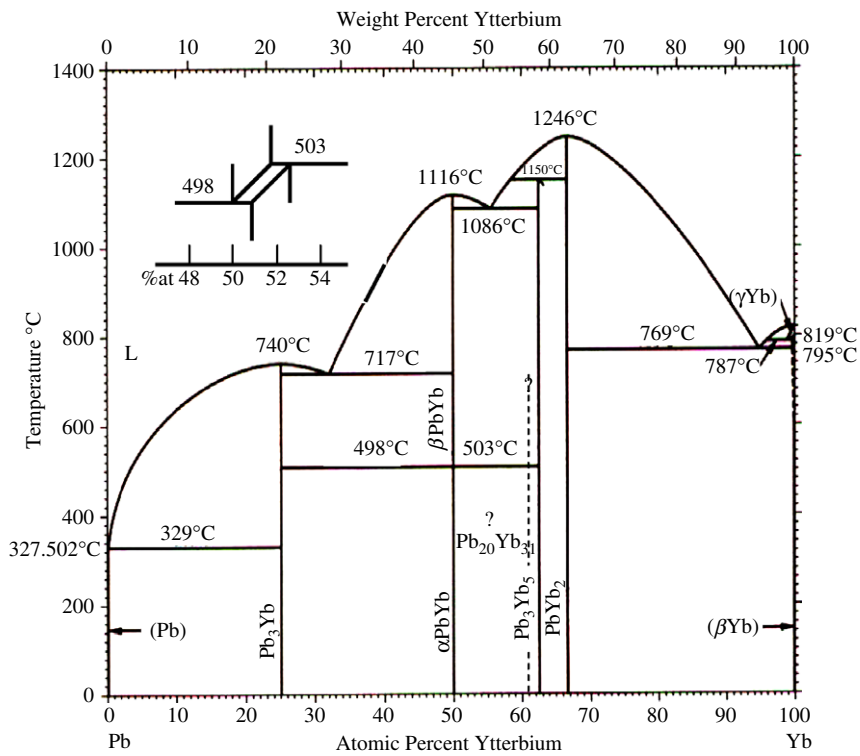

Fig. 1.

where H_i^{SER} (298.15 K) is the molar enthalpy of element i at 298.15 K in its standard element reference (SER) state, FCC_A1 for Pb and Yb.

In this paper, the Gibbs energy functions are taken from the SGTE compilation of Dinsdale [10].

3.2. Liquid phase

The liquid phase has been described by the association model by Sommer [11]. The association model is based on the hypothesis that a C complex is formed in the liquid phase (where A and B are pure elements initially present in the mixing) as follows:

$pA + qB \Leftrightarrow A_pB_q$ (A_pB_q corresponds to the C complex with p and q the stoichiometric coefficients). Such a formation of a C complex in the liquid phase can be suggested by the experimental enthalpies of formation of the intermetallic phases which show a nearly triangular shaped concentration dependence. As in the Ba–Pb [1], and Ca–Pb [2] systems, the compound which presents the highest melting point and enthalpy of formation has been taken as the associated complex. In agreement with the description and calculations made by Sommer [8,11], $PbYb_2$ is the most probable liquid associated complex (C).

Therefore, the Gibbs energy of one mole of formula unit is expressed as the sum of four terms:

$$G^{Liq} - H^{SER} = {}^{ref}G^{Liq} + {}^{form}G_C^{Liq} + {}^{id}G^{Liq} + {}^{ex}G^{Liq}. \quad (2)$$

In a previous paper [12], we briefly gave the equivalence between the formalism described by Sommer [11] and the different terms of Eq. (2) involved in the Thermo-Calc software.

$$\begin{aligned} {}^{ref}G^{Liq} &= \{x_{Pb}[{}^0G_{Pb}^{Liq}(T) - H_{Pb}^{SER}] + x_{Yb}[{}^0G_{Yb}^{Liq}(T) - H_{Yb}^{SER}]\} \\ &\quad \times (y_{Pb} + (p+q)y_C + y_{Yb}) \quad (3) \\ {}^{ref}G^{Liq} &= \{x_{Pb}[{}^0G_{Pb}^{Liq}(T) - H_{Pb}^{SER}] + x_{Yb}[{}^0G_{Yb}^{Liq}(T) - H_{Yb}^{SER}]\}(1 + 2y_C) \end{aligned}$$

$${}^fG_C^{Liq} = (y_{Pb}\,{}^0G_{Pb}^{Liq}(T)) + (y_{Yb}\,{}^0G_{Yb}^{Liq}(T)) + (y_C\,{}^0G_C^{Liq}(T)) \quad (4)$$

$${}^{id}G^{Liq} = RT(y_{Pb}\ln y_{Pb} + y_{Yb}\ln y_{Yb} + y_C\ln y_C) \quad (5)$$

where R is the perfect gas constant; T is the temperature.

The three species A, B, C in the liquid phase have y_A, y_B and y_C as mole fractions, normalized to $y_A + y_B + y_C = 1$. x_A and x_B are the absolute mole fractions of the two components A and B of the liquid, without considering the associated species, normalized to $x_B + x_A = 1$.

For a regular liquid solution, $C^{\mathrm{reg}}_{\mathrm{Pb,Yb}}$ in [8] is equal to the ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Pb,Yb}}$ term of the Redlich–Kister polynomial [13] of the ${}^{ex}G^{\mathrm{Liq}}$ energy part in Eq. (2):

$$ {}^{ex}G_{\mathrm{liq}} = y_i y_j \sum_{v=0}^{v} L^{\mathrm{Liq}}_{i,j}(y_i - y_j)^{v} \tag{6} $$

where i and j are indices which correspond to the three species Pb, Yb and Pb_1Yb_2.

The binary interaction parameters of the ${}^{0}L^{\mathrm{Liq}}_{i,j}$ type, evaluated in the present work, were temperature dependent as follows:

$$ {}^{0}L^{\mathrm{Liq}}_{i,j} = a_0 + b_0 T; \tag{7} $$

a_i and b_i are the coefficients to be optimized.

3.3. Stoichiometric compounds

The Gibbs energy of the stoichiometric compounds (Pb_3Yb, Pb_3Yb_5 and $PbYb_2$) denoted as ${}^{0}G_{\mathrm{ApBq}}$ is expressed as follows:

$$ {}^{0}G_{\mathrm{ApBq}} = p\,{}^{0}G_A + q\,{}^{0}G_B + a + bT \tag{8} $$

where ${}^{0}G_A$ and ${}^{0}G_B$ are the Gibbs energy of the pure elements Pb and Yb respectively; a and b are parameters to be determined. In the case of $PbYb_2$ the best optimization of the binary Pb–Yb system was obtained with its Gibbs energy expressed as a power series: ${}^{0}G_{\mathrm{PbYb_2}} = a + bT + cLn(T) + dT^{-1} + eT^3 + fT^2$.

3.4. Solution phases

The solution phases (FCC_A1 and BCC_A2) were modelled as substitutional solutions. The Gibbs energy of one mole of formula unit of the phase ϕ is expressed as the sum of the reference part ${}^{\mathrm{ref}}G^{\phi}$, the ideal part ${}^{\mathrm{id}}G^{\phi}$, and the excess part ${}^{\mathrm{xs}}G^{\phi}$:

$$ G^{\phi}_m = {}^{\mathrm{ref}}G^{\phi} + {}^{\mathrm{id}}G^{\phi} + {}^{\mathrm{xs}}G^{\phi} \tag{9} $$

As used in the Thermo-Calc software [14],

$$ \begin{aligned} {}^{\mathrm{ref}}G^{\phi}(T) \;=\; & ({}^{0}G^{\phi}_{\mathrm{Pb}}(T) - H^{\mathrm{SER}}_{\mathrm{Pb}}(298.15))x_{\mathrm{Pb}} \\ & + ({}^{0}G^{\phi}_{\mathrm{Yb}}(T) - H^{\mathrm{SER}}_{\mathrm{Yb}}(298.15))x_{\mathrm{Yb}} \end{aligned} \tag{10} $$

$$ {}^{\mathrm{id}}G^{\phi} = RT(x_{\mathrm{Pb}} \ln x_{\mathrm{Pb}} + x_{\mathrm{Yb}} \ln x_{\mathrm{Yb}}) \tag{11} $$

where R and T have been defined above; x_{Pb} and x_{Yb} are the mole fractions of the elements Pb and Yb, respectively.

The excess terms of all the solution phases were modelled by the Redlich-Kister [13] formula.

$$ \begin{aligned} {}^{\mathrm{xs}}G^{\phi}_m(T) \;=\; & x_{\mathrm{Pb}}x_{\mathrm{Yb}}({}^{0}L^{\phi}_{\mathrm{Pb,Yb}}(T) + {}^{1}L^{\phi}_{\mathrm{Pb,Yb}}(T)(x_{\mathrm{Pb}} - x_{\mathrm{Yb}}) \\ & + {}^{2}L^{\phi}_{\mathrm{Pb,Yb}}(T)(x_{\mathrm{Pb}} - x_{\mathrm{Yb}})^2 + \cdots) \end{aligned} \tag{12} $$

with

$$ {}^{i}L^{\phi}_{\mathrm{Pb,Yb}}(T) = a_i + b_i T \tag{13} $$

where ${}^{i}L^{\phi}_{\mathrm{Pb,Yb}}(T)$ is the ith interaction parameter between the elements Pb and Yb, which is evaluated in the presented work. The a_i and b_i coefficients are optimized using the PARROT module [15].

3.5. Intermediate phases with homogeneity ranges

The αPbYb_LT and βPbYb_HT Pb sub-stoichiometric intermetallic compounds, which have a homogeneity range, were treated by a two-sublattice model as $(\mathrm{Pb\%, Yb})_{0.5}(\mathrm{Yb})_{0.5}$. The symbol % denotes the major component in the first sublattice. The Gibbs energy function per mole (m) of the formula unit $(\mathrm{Pb\%, Yb})_{0.5}(\mathrm{Yb})_{0.5}$ is the following:

$$ \begin{aligned} & G^{0.5\mathrm{Pb}0.5\mathrm{Yb}} - H^{\mathrm{SER}}_{0.5\mathrm{Pb}0.5\mathrm{Yb}} = y^1_{\mathrm{Pb}}y^2_{\mathrm{Yb}}\,{}^{0}G^{PbYb} + y^1_{\mathrm{Yb}}y^2_{\mathrm{Yb}}\,{}^{0}G^{\mathrm{YbYb}} \\ & + RT\left[0.5(y^1_{\mathrm{Pb}}Lny^1_{\mathrm{Pb}} + y^1_{\mathrm{Yb}}Lny^1_{\mathrm{Yb}})\right] + {}^{\mathrm{xs}}G^{\lambda}_m \end{aligned} \tag{14} $$

where $y^2_{\mathrm{Yb}} = 1$ denotes the site fraction of ytterbium in the second sublattice, y^1_{Pb} and y^1_{Yb} the site fractions of lead and ytterbium in the first sublattice, and ${}^{0}G^{\mathrm{Yb}_{0.5}\mathrm{Yb}_{0.5}}$ is the Gibbs energy of the hypothetical compounds $Yb_{0.5}Yb_{0.5}$. ${}^{0}G^{\mathrm{Pb}_{0.5}\mathrm{Yb}_{0.5}}$ is the Gibbs energy of the stoichiometric compound $Pb_{0.5}Yb_{0.5}$.

$$ H^{\mathrm{SER}}_{0.5\mathrm{Pb}0.5\mathrm{Yb}} = 0.5H^{\mathrm{SER}}_{\mathrm{Pb}} + 0.5H^{\mathrm{SER}}_{\mathrm{Yb}}. \tag{15} $$

${}^{\mathrm{xs}}G^{\lambda}_m$ is the excess Gibbs energy expressed by the following expression:

$$ {}^{\mathrm{xs}}G^{\lambda}_m = y^1_{\mathrm{Pb}}y^1_{\mathrm{Yb}}(L^{\lambda}_{\mathrm{Pb,Yb:Yb}}) \tag{16} $$

where $L^{\lambda}_{\mathrm{Pb,Yb:Yb}}$ represents the interaction parameters between the elements Yb and Pb in the first sublattice while the second sublattice is only occupied by the element Yb. These excess parameters are temperature dependent:

$$ {}^{\lambda}a + {}^{\lambda}b\,T. \tag{17} $$

In order to avoid the occurrence of the hypothetical compound $Yb_{0.5}Yb_{0.5}$ during the phase diagram calculation, the value + 5000 J/mol of atoms was added to GHSERYb; see Table 2.

4. Results and discussions

The optimizations of the interaction parameters were carried out by taking into consideration all the relevant experimental data which are detailed in Section 2. But, in order to avoid the occurrence of an unwanted inverted miscibility gap in the liquid phase during the calculation of the phase diagram, additional constraints $\partial^2G/\partial^2x > 0$ in the whole Yb composition range and at different temperatures between 1500 and 5000 K were imposed. The modelling of the system was performed according to the CALPHAD method [3] using the Thermo-Calc software [14] and the assessment was carried out by using the PARROT module [15]. The optimized thermodynamic description of the Pb–Yb system obtained in the present work is shown in Table 2. The $\Delta H^{0}_{\mathrm{Pb_1Yb_2}}$ optimized enthalpy of formation of the associate ($-207\,606$ J (mol at.)$^{-1}$) in the liquid phase is in good agreement with the one by Sommer et al. ($-203\,000$ J (mol at.)$^{-1}$) [8] but its $\Delta S^{0}_{\mathrm{Pb_1Yb_2}}$ optimized entropy of formation (-93.7 J (mol at.)$^{-1}$ K^{-1}) is slightly higher than the one (-68.2 J (mol at.)$^{-1}$) determined by this author. First, the optimization of the parameters of the liquid phase was carried out with only one excess parameter ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Pb,Yb}}$, but to give account of all the thermodynamic data of this phase, it was necessary to optimize two extra excess terms ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Pb,Pb1Yb2}}$ and ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Pb1Yb2,Yb}}$ which describe the interaction of the free atoms Pb and Yb with the Pb_1Yb_2 associates. The optimized value ($-99\,046$ J (mol at.)$^{-1}$) of the excess parameter of the regular liquid solution, ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Pb,Yb}}$, is different from the one ($-27\,300$ J (mol at.)$^{-1}$) calculated by [8], which is normal due to the involved extra excess terms. The computed phase diagram (Figs. 2 and 3) is very close to the experimental one (Fig. 1). The invariant equilibria of Pb–Yb system are listed in Table 3. As shown in this table, a very good agreement is obtained between the calculations and the experiments. The assessed enthalpy of mixing of liquid Pb–Yb alloys referring to liquid components at $T = 1060$ K and $T = 1100$ K are plotted in Fig. 4. An asymmetric composition dependence is observed which indicates the presence of short-range order in the liquid phase described by the formation of associates. Also shown in this figure are the data from [8]. A satisfactory agreement is observed. The enthalpies of formation of the Pb–Yb intermetallic compounds (measured and calculated) are collected in Table 4, also are presented in Fig. 5. A satisfactory agreement is noted. Concerning the optimized

Table 2
The optimized thermodynamic parameters of the Pb–Yb system.

Phase	Thermodynamic model	Parameters (units in J/mol at. and J/mol at. K)
Liquid	$(Pb, Yb, Pb_1Yb_2)_1$	${}^0G^{Liq}_{Pb_1Yb_2} - {}^{298}H^{FCC_A1}_{Pb} - 2\,{}^{298}H^{FCC_A1}_{Yb} = -207606.771 + 93.70182T + GLIQPb + 2GLIQYb$ ${}^0L^{Liq}_{Pb,Yb} = -99046.4539 - 7.58962T$ ${}^0L^{Liq}_{Pb,Pb1Yb2} = -4978.92351 - 131.29188T$ ${}^0L^{Liq}_{Pb1Yb2,Yb} = -24616.4052 + 14.68614T$
BCC_A2	$(Pb, Yb)_1(Va)_3$[a]	${}^0L^{BCC_A2} = -149553.158 + 28.513725T$ ${}^1L^{BCC_A2} = -14740.1227 + 1.5T$
FCC_A1	$(Pb, Yb)_1(Va)_1$	${}^0L^{FCC_A1} = -116698.561 + 3.96519T$ ${}^1L^{FCC_A1} = -1400 + T$
Pb_3Yb	$(Pb)_{0.75}$: $(Yb)_{0.25}$	$G^{Pb_3Yb}_{Pb:Yb} - 0.75\,{}^{298}H^{FCC_A1}_{Pb} - 0.25\,{}^{298}H^{FCC_A1}_{Yb} = -38628.9628 + 45.3525127T - 15.8254777TLn(T) - 0.106845214T^{-1} + 0.000002360833 85T^3 - 0.0115568711T^2$
αPbYb	$(Pb)_{0.5}$: $(Yb)_{0.5}$	$G^{\alpha PbYb}_{Pb:Yb} - 0.5\,{}^{298}H^{FCC_A1}_{Pb} - 0.5\,{}^{298}H^{FCC_A1}_{Yb} = 0.5\,{}^{298}G^{FCC_A1}_{Pb} + 0.5\,{}^{298}G^{FCC_A1}_{Yb} - 63142.517 + 8.14968T$ $G^{\alpha PbYb}_{Yb:Yb} = {}^{298}G^{FCC_A1}_{Yb} + 5000$ ${}^0L_{Pb,Yb:Yb} = -44000 + 19T$
βPbYb	$(Pb)_{0.5}$: $(Yb)_{0.5}$	$G^{\beta PbYb}_{Pb:Yb} - 0.5\,{}^{298}H^{FCC_A1}_{Pb} - 0.5\,{}^{298}H^{FCC_A1}_{Yb} = 0.5\,{}^{298}G^{FCC_A1}_{Pb} + 0.5\,{}^{298}G^{FCC_A1}_{Yb} - 63095.2883 + 8.08835T$ $G^{\beta PbYb}_{Yb:Yb} = {}^{298}G^{FCC_A1}_{Yb} + 5000$ ${}^0L_{Pb,Yb:Yb} = -24756.7541 + 5.89201T$
Pb_3Yb_5	$(Pb)_{0.375}$:$(Yb)_{0.625}$	$G^{Pb_3Yb}_{Pb:Yb} - 0.375\,{}^{298}H^{FCC_A1}_{Pb} - 0.625\,{}^{298}H^{FCC_A1}_{Yb} = 0.375\,{}^{298}G^{FCC_A1}_{Pb} + 0.625\,{}^{298}G^{FCC_A1}_{Yb} - 68078.7434 + 15.15153T$
$PbYb_2$	$(Pb)_{0.333}$: $(Yb)_{0.667}$	$G^{Pb_3Yb}_{Pb:Yb} - 0.333\,{}^{298}H^{FCC_A1}_{Pb} - 0.667\,{}^{298}H^{FCC_A1}_{Yb} = -64005.8791 + 15.2122638T - 9.45857772TLn(T) + 112.576608T^{-1} + 0.000001528544 36T^3 - 0.0138979842T^2$

[a] Va stands for vacancy.

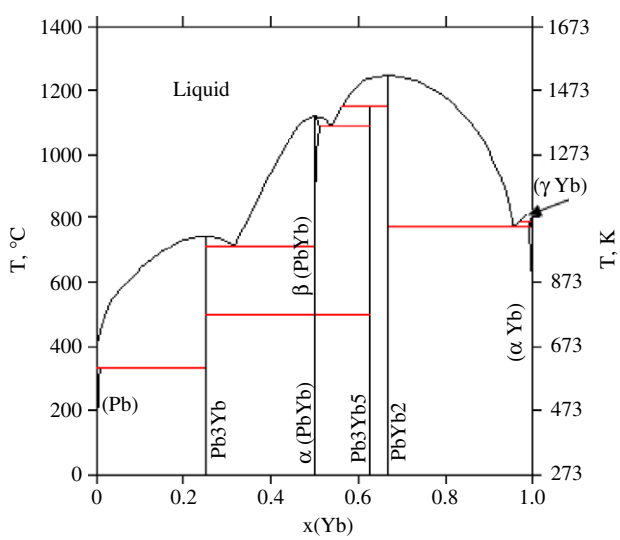

Fig. 2. Calculated Pb–Yb phase diagram (this work).

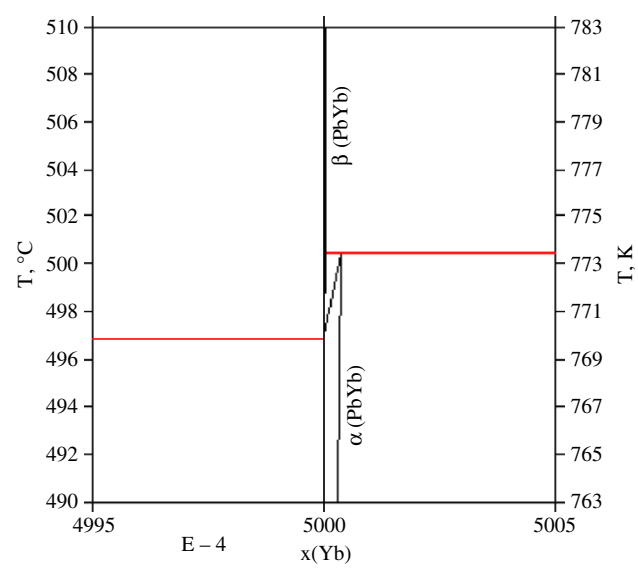

Fig. 3. Enlarged section of the Pb–Yb phase diagram around $x(Yb) = 0.5$ at.

melting enthalpy of Pb_3Yb (16.29 kJ/mol at.), it is in good agreement with the experimental data by [8] (15.5 ± 1.2 kJ/mol at. measured by solution calorimetry or 14.0 ± 2 kJ/mol determined by dynamic differential calorimetry). In contrast, the agreement between the calculated melting enthalpies of $PbYb_2$ (32 kJ/mol at.) and of βPbYb_HT (19.8 kJ/mol at.) and the experimental data [8] (respectively 10.2 ± 1.5 kJ/mol at. and 10 ± 2 kJ/mol at. both determined by solution calorimetry) are less good.

Fig. 6 shows the calculated molar heat capacity of $PbYb_2$, Pb_3Yb and PbYb as a function of temperature with data from the literature [5]. The calculated values for $PbYb_2$ and Pb_3Yb are in good agreement with the experimental results from [5] but the optimized molar heat capacity of PbYb is somewhat higher than the one determined in [5]. The singular point at 583 K is corresponding to the joining temperature of two thermodynamic descriptions of pure element (Pb and Yb) for different temperature ranges in [10].

As first stated by [16], we verified that when the liquid phase is suspended, the intermetallic compounds are no more stable at high

Table 3
Invariant reactions in the Pb–Yb System.

Reaction	McMasters and Gschneidner [4]			This work		
	T (K)	x (Sn)		T (K)	x (Sn)	
Liq ↔ (Pb)+Pb_3Yb	602	Liq (Pb)	<0.001 0.0015	606.3	Liq (Pb)	0.00073 0.008
Liq ↔Pb_3Yb	1013			1015		
Liq ↔Pb_3Yb+ βPbYb	990	Liq	0.32	984.1	Liq	0.316
Liq ↔βPbYb	1389			1391		
αPbYb βPbYb	778			773.6		
Liq ↔Pb_3Yb_5 + βPbYb	1359	Liq βPbYb	0.55 0.52	1360	Liq βPbYb	0.54 0.51
Liq + $PbYb_2$ Pb_3Yb_5	1423	Liq	0.58	1423	Liq	0.56
Liq ↔$PbYb_2$	1519			1518		
Liq ↔ (βYb)+ $PbYb_2$	1042	Liq (βYb)	0.949 0.998	1044	Liq (βYb)	0.954 0.989
Liq + (βYb) ↔ (γYb)	1060	Liq (βYb) (γYb)	0.96 0.998 0.99	1058	Liq (βYb) (γYb)	0.96 0.998 0.99

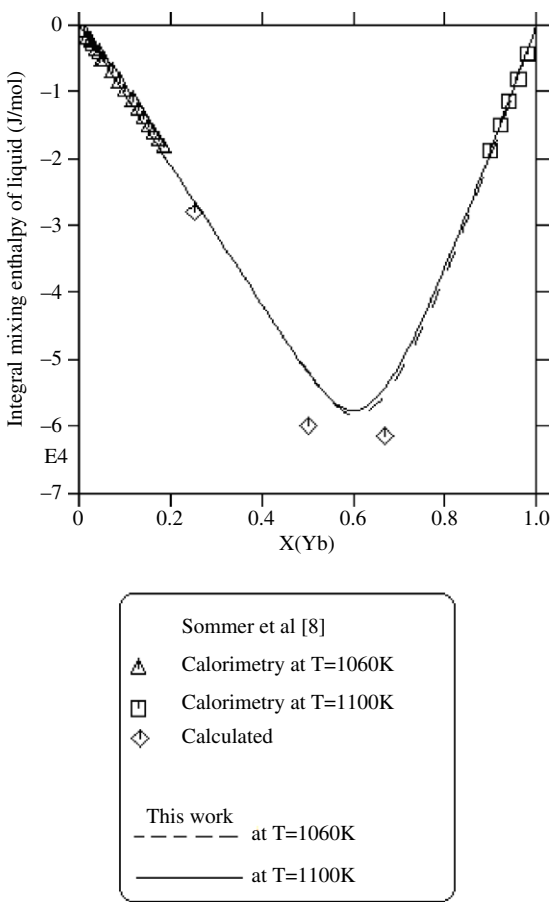

Fig. 4. Comparison of calculated and measured integral enthalpy of mixing of the liquid phase at $T = 1060$ K and $T = 1100$ K, with reference to the both liquid components.

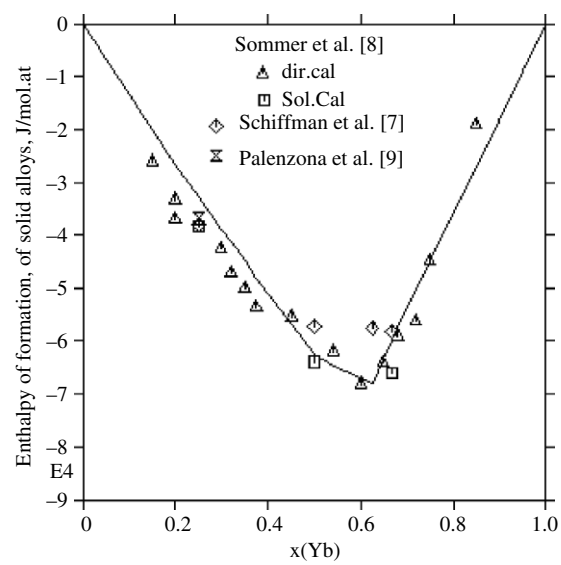

Fig. 5. Calculated and measured enthalpies of formation of the intermetallic compounds.

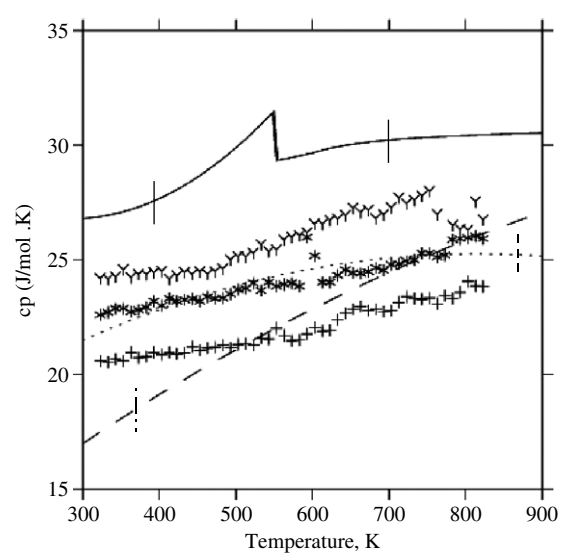

Fig. 6. Calculated and measured molar heat capacity of Pb_3Yb (∗), PbYb (+) and $PbYb_2$ (Y) (J/mol at. K). The error bar was estimated to be 2 J/mol at. K.

temperatures and disappear at lower temperatures; see Fig. 7. At high temperatures, the BCC_A2 terminal solid solution is stable on the Yb-rich side, and the FCC_A1 terminal solid solution is stable on the Pb-rich side; between these single phase domains the two-phase domain (BCC_A2 + FCC_A1) is well calculated.

5. Conclusion

A consistent set of thermodynamic parameters of the different phases of the Pb–Yb binary system has been optimized. The new phase $Pb_{20}Yb_{31}$ identified by [6] was not taken into consideration because only its crystal structure has been determined and not its invariant reactions. The liquid phase was described with a thermodynamic model including the presence of a chemical short-range

Table 4
Calculated and measured enthalpies of formation of intermetallic compounds.

Phase	Enthalpy of formation kJ/mol	Technique used	Reference
$Pb\,Yb_2$	−66±5	Solution calorimeter	[8]
	−67±3	Direct calorimeter	[8]
	−58.2±3	Knudsen data	[7]
	−59.99	Optimization	This work
Pb_3Yb_5	−55.5±2	Direct calorimeter	[8]
	−57.7±1	Knudsen data	[7]
	−66.5±2	Direct calorimeter	[8]
	−68.02	Optimization	This work
αPbYb	−62.59	Optimization	This work
βPbYb	−64±2.5	Solution calorimeter	[8]
	−65±2	Direct calorimeter	[8]
	−57±1	Knudsen data	[7]
	−62.16	Optimization	This work
Pb_3Yb	−38.5±3	Solution calorimeter	[8]
	−39±1	Direct calorimeter	[8]
	−38.1±1	Knudsen data	[7]
	−37±2	Dynamic differential calorimeter data	[9]
	−32.96	Optimization	This work
Phase	**Enthalpy of melting kJ/mol**		
$Pb\,Yb_2$	10.2±1.5	Solution calorimeter	[8]
	32	Optimization	This work
βPbYb	10±2	Solution calorimeter	[8]
	19.8	Optimization	This work
Pb_3Yb	15.5±1.2	Solution calorimeter	[8]
	14±2	Dynamic differential calorimeter data	[9]
	16.29	Optimization	This work

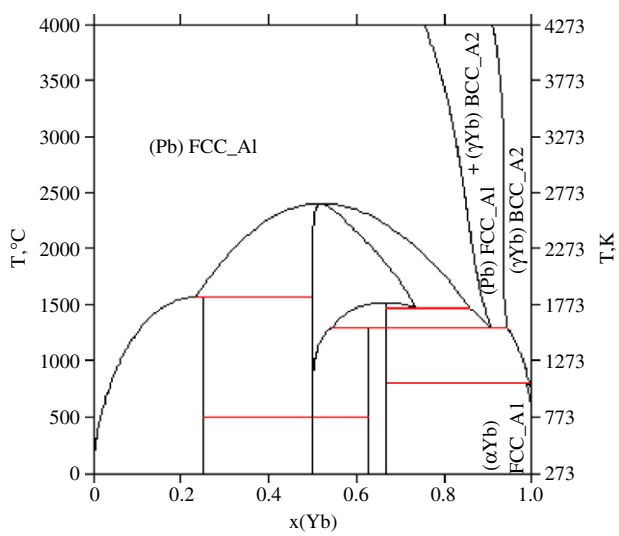

Fig. 7. Calculated Pb–Yb phase diagram when the liquid phase is suspended.

order due to the formation of the Pb_1Yb_2 stoichiometric associated complex. A small homogeneity range on the Yb-rich side has been optimized for αPbYb_BT and βPbYb_HT but it is less important than the experimental one. The computed values are in good agreement with the experimental thermodynamic (except for the molar heat capacity of $PbYb_2$) and phase diagram data and we verified that no unwanted inverted miscibility gap was calculated for the solution phases up to 5000 K.

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